



Sir Syed
C@SE Institute of Technology, Islamabad
Department of Computer Science

Assignment 5

Due Date: May 26, 2026

Course Title:	Web Engineering	Course Code	CSC3008	Credit Hours:	3(2,1)
Instructor:	Mr. Muhammad Hassan Shah	Program Name:	BS (CYS/CS)		

Instructions for Students:

- Do not copy code directly — understand and implement
- Keep your code clean and well-structured
- Focus on quality over quantity
- Late submissions may result in mark deductions

Title: React

[CLO2, CLO3]

Q1. Functional Component & JSX Create a React functional component called StudentCard that accepts name, rollNumber, and grade as props and displays them in a styled card layout.

Q2. useState Hook Build a simple counter app with three buttons — Increment, Decrement, and Reset. The counter should never go below 0. Display the current count prominently on screen.

Q3. Rendering Lists Given an array of 5 products (name, price, category), render them as a list using .map(). Each item must have a unique key prop. Add a heading that shows the total number of products.

Q4. Controlled Form with Validation Build a registration form with fields: Name, Email, Password, and Confirm Password. Use useState to control all inputs and validate that passwords match before showing a success message.

Q5. useEffect Hook Create a component that fetches a random joke from a public API (https://official-joke-api.appspot.com/random_joke) when the page loads and displays it. Add a "New Joke" button that fetches another joke on click.

Q6. Conditional Rendering Build a simple login/logout toggle. If the user is "logged in", show a welcome message with their name and a Logout button. If logged out, show a Login button. Manage state using useState.

Q7. Component Composition & Lifting State Up Create a temperature converter app with two child components — one for Celsius input and one for Fahrenheit input. Lift the state up to the parent so both inputs stay in sync when either one is changed.

Q8. useContext for Global State Build a theme switcher (Light/Dark mode) using the React Context API. The theme toggle button should be in a Navbar component, and the background/text color of the entire app should update globally without prop drilling.

Q9. Custom Hook Write a custom hook called useFetch(url) that handles loading, error, and data states for any API call. Use this hook in at least two different components to fetch data from different endpoints.

Q10. React Router with Dynamic Routes Build a multi-page app with React Router that includes:

- A Home page with a list of 5 blog posts
- A Post Detail page at /posts/:id that shows full details of the clicked post
- A 404 Not Found page for invalid routes